HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and optimised the PMO programme for a **GAMP 5 compliant aseptic drug product facility**, integrating **digital tools** to enhance maintenance predictability and target specific asset needs.
- Analysed equipment performance data, including **downtime trends and MTBF/MTTR metrics**, to identify chronic issues and implement sustainable **reliability improvements** across critical plant machinery.
- Collaborated with system owners, planners, and technicians to define maintenance scopes, refine job plans, and ensure maintenance tasks were **risk-based and executable**, supporting shutdown activities.
- Supported **Drug Product manufacturing** within Vial and Syringe filling plants, ensuring adherence to stringent **GxP** and EHS requirements.
- Employed business analytics to assess performance and identify improvement opportunities for **Equipment Reliability**, Maintenance Reliable Execution, and **Predictive Maintenance strategies**.
- Drove continuous improvement initiatives by capturing lessons learned and embedding **standard work and technical documentation** into future maintenance execution cycles.
- Managed system improvements and developed detailed specifications and **standard operating procedures (SOPs)** for plant equipment, contributing to operational excellence.

SKILLS

Engineering & Reliability – Maintenance PMO, Reliability Framework, Equipment Reliability, Predictive Maintenance, Preventive Maintenance, Asset Lifecycle Performance, Criticality Analysis, FMEA, RCA, TPM, Lean, Six Sigma, CMMS/EAM Systems

Manufacturing & Compliance – GxP, EHS, Regulatory Compliance, Aseptic Processing, Pharmaceutical Manufacturing, Biotechnology Manufacturing, Drug Product Manufacturing, Vial Filling, Syringe Filling, Lyophilisation

Data Analysis & Tools – Business Analytics, Data Analysis, Downtime Analysis, Failure Trend Analysis, MTBF, MTTR, SQL, Python Scripting, Splunk, Dynatrace, CloudWatch

Software & Systems – Java Spring Boot, REST APIs, Python, AWS, Docker, Linux, Jenkins, Jira, Confluence, ERP Systems, Enterprise Business Systems

Analytics & Visualisation – Power BI, Tableau, DAX, Power Query, Data Modelling, Data Warehousing, R / RStudio, pandas, NumPy, scikit-learn

PROJECTS

Automated Maintenance Scheduling System (HCL Technologies)

- Developed a Python-based system to automate the generation of preventative maintenance schedules for enterprise clients, integrating with existing **CMMS** databases.
- Designed and implemented SQL queries across **PostgreSQL** and Oracle to extract asset data, analyse maintenance history, and predict optimal scheduling intervals, improving resource allocation efficiency by **18%**.
- Created automated reports using Python scripting to provide stakeholders with insights into maintenance compliance and equipment uptime, facilitating better operational decision-making.

Production Equipment Performance Monitoring Dashboard (MSc Applied Project)

- Built an interactive **Power BI** dashboard to visualise key performance indicators for manufacturing equipment, including uptime, downtime causes, and mean time between failures (MTBF).
- Integrated data from disparate sources using **Power Query** and modelled relationships in Power BI to provide a unified view of asset health for business stakeholders.
- Applied statistical analysis techniques in R to identify correlations between maintenance activities and equipment reliability, informing strategy for proactive maintenance.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)